

DUC THINH LE

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EDUCATION

Hanoi University of Science and Technology (HUST), Vietnam

Dec. 2021 –

Master of Science in Control Engineering and Automation

Nov. 2023

- CPA: 3.72/4.00 (with Distinction)
- Thesis: [Nonlinear Observer-based Motion Control of Electric Vehicles](#)

Hanoi University of Science and Technology, Vietnam

Sep. 2016 –

Engineer in Control Engineering and Automation

Aug. 2021

- Thesis: [Disturbance Observer-based Backstepping for Electric Vehicles Motion Control](#)

PROFESSIONAL EXPERIENCES

Lecturer, Thuyloi University, Vietnam

2024 – Present

Grant writing & Project management

- Developed early-stage research ideas into fundable proposal concepts for two approved grants: [an industry-funded VINIF project](#) on motion control of multi-unmanned vehicles (**\$152K; started Dec. 2025**) and a **publicly funded NAFOSTED national research grant** on safety control for multi-shaft systems (**\$68K; started Apr. 2025**).
- Prepared full grant application packages, including proposal narratives, technical work plans, itemized budgets, vendor quotations, and supporting documents for external funding evaluation.
- Served as **scientific secretary** after project approval, coordinating team communication, tracking milestones and deliverables, organizing technical documents, and supporting PI-level progress reporting.
- Acted as a **key research team member** during project implementation, developing technical ideas, conducting literature review and method design, drafting manuscripts, and preparing responses to peer-review comments.

Research & Technical Development

- Published **15 peer-reviewed international journal papers** over a two-year period, including ISA Transactions, IEEE Access, and International Journal of Robust and Nonlinear Control, on motion control for electromechanical systems.
- Inventor of **two pending Vietnamese patent application**: Centrifugal Air Compressor Driven by an Electric Motor using Conical Active Magnetic Bearings (Application [No. 1-2025-08826](#)) and Self-Balancing Electric Wheelchair System integrated with Piezoelectric Sensors.
- Developed fault-tolerant control strategies for nonlinear systems (such as **electric drives, electric cars, electromechanical systems, and renewable energy**) with input saturation and time delays, ensuring prescribed-time convergence, bounded tracking errors, and robustness to actuator faults and disturbances.

Teaching

- Courses: Motion Control, Applied Programming, Actuators and Control, Swarm Robotics and Control.

Research Assistant, HUST

2021 – 2025

Research Project Execution

- Prepared grant proposals and administrative documentation for **three university-level research projects**.
- Managed project close-out submissions, including technical reports, compliance files, acceptance documents, and final project dossiers.

Research & Technical Contributions

- Designed advanced controllers, including fractional-order, nonlinear, and adaptive controllers, across projects on roll-to-roll systems (2021-2022), electric vehicles (2022-2023), and non-contact drives (2024-2025).
- Developed mathematical models and validated control algorithms through MIL simulation using PLECS and CarSim, PIL testing using MATLAB and TI C2000 MCUs, and HIL testing using Typhoon HIL and dSPACE.
- Authored and reviewed research manuscripts in control systems, motion control, and electromechanical systems.

SELECTED PUBLICATIONS

- [1] **Le, D. T.**, Nguyen, D. G., Ta, T. T., Nguyen, T. L., Nhat, M. V., & Nguyen, D. H. (2025). Fractional Order Approach for Conical Active Magnetic Bearings Under Time Delay and Supply Faults. *IEEE Access*, vol. 13, pp. 119157-119176.
- [2] **Le, D. T.**, Nguyen, B. M., Nguyen, T. L., & Hoang, D. C. (2024). An Axial Gap Bearingless Motor Drive System with Non-linear Disturbance Observers-Based Robust Control. *Journal of Control, Automation and Electrical Systems*, 35(1), 180-190.
- [3] **Le, D. T.**, Ngo, M. T., & Nguyen, T. L. (2023). A sensorless anti-windup speed control approach to axial gap bearingless motors with nonlinear lumped mismatched disturbance observers. *ISA transactions*, 138, 408-431.
- [4] **Le, D. T.**, Nguyen, D. T., Le, N. D., & Nguyen, T. L. (2022). Traction control based on wheel slip tracking of a quarter-vehicle model with high-gain observers. *International Journal of Dynamics and Control*, 10(4), 1130-1137.
- [5] Pham, X. D., Nguyen, T. L., & **Le, D. T.** (2026). Fixed-time fault-tolerant control for electro-mechanical braking systems under input saturation and time-delay. *ISA transactions*.

- [6] Nguyen, T. D., Nguyen, T. L., & **Le, D. T.** (2026). Robust Prescribed Performance Control for Heavy-Haul Freight Trains Under Actuator Faults and Input Time Delays. *Journal of Dynamic Systems, Measurement, and Control*, 1-38.
- [7] Nguyen, T. D., Nguyen, T. L., & **Le, D. T.** (2026). Robust predefined-time tracking control with output constraint for autonomous railway vehicles under actuator saturation and input time delays. *European Journal of Control*, 101526.
- [8] Nguyen, H. D., Nguyen, T. L., Vu, M. N., & **Le, D. T.** (2025). Robust Model Predictive Control-based Recurrent Neural Networks for Autonomous Vehicles in Avoidance Collisions. *IEEE Access*, vol. 13, pp. 106115-106128.
- [9] Dung Nguyen, T., **Le, D. T.**, Manh Do, D., & Lam Nguyen, T. (2026). Robust Fixed-Time Sliding Mode with Prescribed Performance Control for Heavy-Haul Freight Trains Under Actuator Saturation and Input Time Delay. *International Journal of Robust and Nonlinear Control*, 36(4), 1838-1864.
- [10] Nguyen, N. T., **Le, D. T.**, Nguyen, B. M., & Nguyen, T. L. (2025). Asymmetric time-varying integral barrier Lyapunov control-based trajectory tracking of autonomous vehicles with input magnitude and rate constraints, *Asian J. Control* 1–20.

HONORS AND AWARDS

Best Master Thesis Award, HUST, Vietnam.	2023
Merit Scholarship for master's students with outstanding academic and research achievements, HUST, Vietnam.	2022
Best Paper Award, 7th National Scientific Conference on Electrical, Electronics, and Automation Engineering, Hanoi University of Industry, Vietnam.	2022
Best Graduation Thesis Award, HUST, Vietnam.	2021

ACTIVITIES

Reviewer	2022 – Present
Reviewer for IEEE Transactions on Transportation Electrification, ISA Transactions, Robotics and Autonomous Systems, Computers and Electrical Engineering, Scientific Reports, International Journal of Robust and Nonlinear Control, Nonlinear Dynamics, Electrical Engineering, Asian journal control.	
Teaching Assistant, HUST	2021 – 2025
Acting as a research team leader, providing academic guidance, and technical support to undergraduate students to achieve:	
- HUST Scientific Research Student Conference: Won 02 First Prizes (2023, 2025) and 01 Second Prize (2024). - Ministry of Education and Training: Won 02 Second Prizes at the "Science and Technology Award for Students in Higher Education Institutions" (2023, 2025).	
Assisted faculty members in teaching undergraduate courses, including Pneumatic and Hydraulic Automation Elements, Electrical Drives, and Graduation Projects.	

SKILLS

Validation & Experimental Methods

- Methodologies: MIL, PIL, HIL
- Simulation environments: MATLAB, PLECS, CarSim, PX4 Autopilot
- Hardware platforms: Typhoon HIL, dSPACE, TI C2000 MCUs, Pixhawk

Programming & Tools

- Languages: MATLAB/Simulink, Python, C/C++, LaTeX

REFERENCES

1. **Assistant Professor [Binh-Minh Nguyen](#)**
 - Assistant Professor at The University of Tokyo, Japan.
 - Research Supervisor
 - Email: nguyen.binhminh@edu.k.u-tokyo.ac.jp
2. **Associate Professor [Tung Lam Nguyen](#)**
 - Head of Motion Control and Applied Robotics Laboratory, HUST, Vietnam
 - Supervisor of the undergraduate and master programs.
 - Email: lam.nguyentung@hust.edu.vn
3. **Associate Professor [Danh Huy Nguyen](#)**
 - Chairman of the Electrical and Electronic Engineering School Council, HUST, Vietnam
 - Co-supervisor of the master programs.
 - Email: huy.nguyendanh@hust.edu.vn